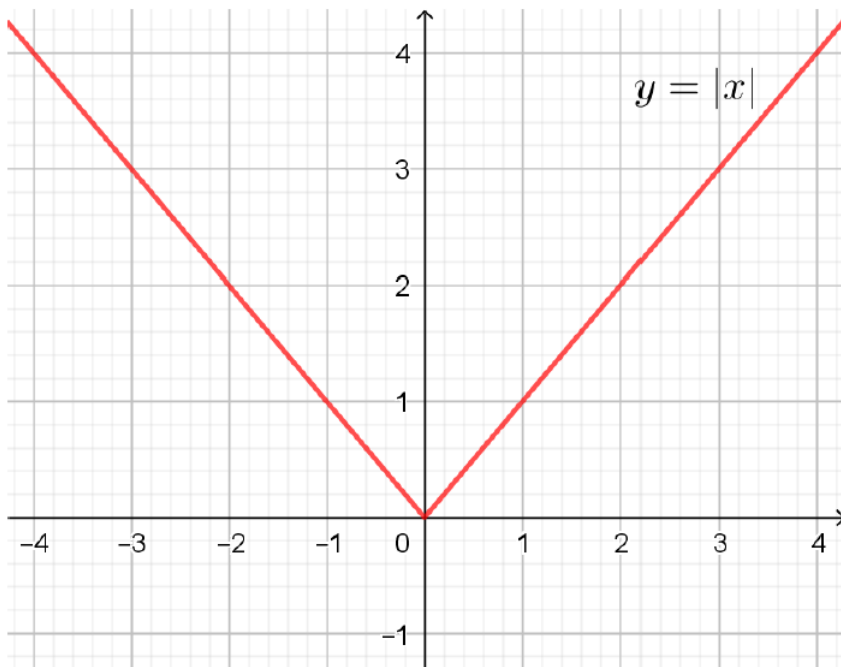


Sec 2.6 – Differentiability

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

Partner up



At $(0, 0)$ the graph of $f(x) = |x|$

- ① has a tangent line at $y = 0$
- ② has infinitely many tangent lines
- ③ has no tangent line
- ④ has two tangent lines $y = -x$ and $y = x$

There is no tangent line, so this function doesn't have a derivative at $x=0$. When a function doesn't have a derivative, we say it is not **differentiable**.

If we zoomed in, the function would still look the same. Sometimes, when zooming in, functions flatten out.

Definition

A function is **differentiable** at x if $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ exists. The value of the limit and the slope of the tangent line are the derivative of f at x .

By the way, this function is called **piecewise linear**. Each part of it is linear. Piecewise functions are defined, wait for it... in pieces...

$$f(x) = |x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Even though we know this function isn't differentiable, let's use our skills in determining limits to try to compute the limit of $f(x) = |x|$ at $x = 0$, in other words we will try to calculate the derivative of $f(x)$ at $x = 0$.

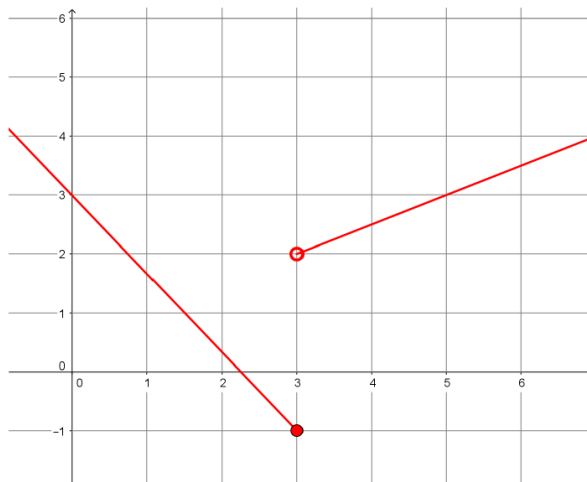
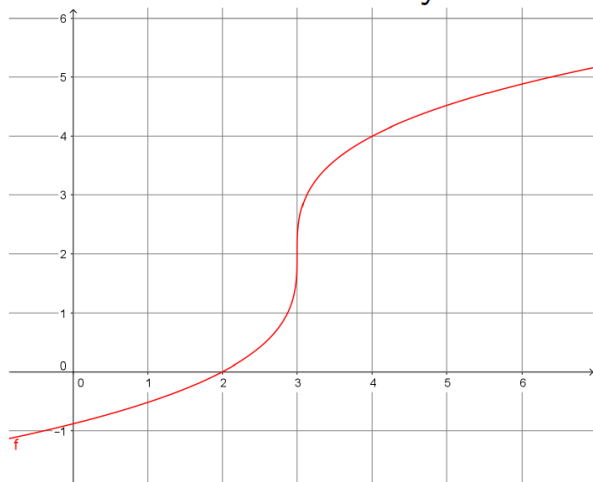
$$\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{|h| - 0}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h}$$

As h approaches 0 from the right, h is positive, so $|h| = h$, and the ratio is always 1. Note that this represents the slope on right side of the graph, which is 1.

As h approaches 0 from the left, h is negative, so $|h| = -h$, and the ratio is always -1. Note that this represents the slope on left side of the graph, which is -1.

Since the limits are different from each side, the limit of the difference quotient does not exist, which is the same as saying the derivative doesn't exist.

Find the value of x for each of the functions below where the function is not differentiable. Why is the function not differentiable?



Three ways derivatives might not exist:

- The function isn't continuous.
- The function has a sharp corner or cusp
- The tangent line is vertical

Suppose $f'(x)$ exists for all x in (a, b) . Which of the following four statements are true?

- $f(x)$ is continuous on (a, b)
- $f(x)$ is continuous at $x = a$
- $f(x)$ is defined for all x in (a, b)
- $f'(x)$ is differentiable on (a, b)

- I and III
- I, II and III
- All of the above
- None of the above
- Only I

(a) is correct.

In II, we can't say anything about the endpoint of the interval, because the derivative is only on (a,b) .

In IV there is an f' , not f .

Theorem (2.1)

If $f(x)$ is differentiable at a point $x = a$, then $f(x)$ is continuous at $x = a$.

Differentiability guarantees continuity, but not the other way around.

If I purchased a Shmuffin, then I was at Sheetz.

I don't purchase a Shmuffin every time I go to Sheetz, but I can only get a Shmuffin at Sheetz.